

Chapter 7

More SQL: Complex Queries, Triggers, Views, and Schema Modification

Outline of the chapter:

- More Complex SQL Retrieval Queries
- Specifying Semantic Constraints as Assertions and Actions as Triggers
- Views (Virtual Tables) in SQL
- Schema Modification in SQL

Comparisons Involving NULL

- What is a NULL?
 - Unknown or unavailable value
- NULL is not a data value! We cannot use equal operator = to check null attributes!
- SQL allows queries that check whether an attribute value is NULL by IS NULL or IS NOT NULL.
- **Query 18.** Retrieve the names of all employees who do not have supervisors.

```
SELECT fname, lname
FROM employee
WHERE employee.super_ssn IS NULL;
```

Nested Queries

- What is a Nested Query?

Using select-from-where statement within WHERE clause of another query.

- Example:

```
SELECT      *
FROM employee
WHERE      dno IN (SELECT      dnumber
                        FROM      department
                        WHERE      dname='Research');
```

- They are also called Correlated Nested Queries.
- To understand nested queries, consider them as two for-loops inside each other;

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For each tuple in outer query, run the inner query and see if there is any result.

To write queries you may need to look at the schema diagram of COMPANY database:

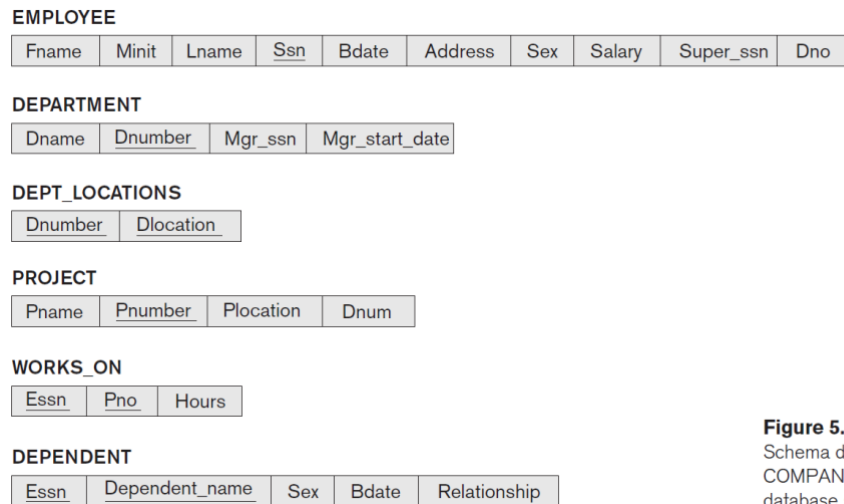


Figure 5.5
Schema diagram for the
COMPANY relational
database schema.

Write a query which prints first name and last name of all department managers (write it as nested queries and with IN operator).

- Can you write a query which only returns managers' SSN?
- In which table are first name and last name?
- How could you know if an employee is a department manager?
- Now compile everything as a nested query:

What the following query will return?

```
SELECT DISTINCT pnumber
FROM project
where pnumber IN
    (SELECT pnumber
     FROM project, department, employee
     where project.dnum= department.dnumber and
           department.mgr_ssn = employee.ssn and
           employee.lname = 'Smith')
OR
pnumber IN
    (SELECT pno
     FROM works_on, employee
     WHERE works_on.essn = employee.ssn and
           employee.lname = 'Smith');
```

Use a tuple of values in comparisons, by placing them within parentheses.

```
SELECT DISTINCT essn
FROM works_on
where (pno, hours) IN (SELECT pno, hours
                       FROM works_on
                       where essn = '123456789');
```

- What does the query return?

Comparing a value to a set/multiset

Operators:

IN

= ANY (or = SOME) operator

$v = ANY V$

Returns TRUE if the value v is equal to some value in the set V .

It is equivalent to IN.

Operators $>$, $>=$, $<$, $<=$, and $<>$ could also be combined with ANY (or SOME).

ANY could be replaced with ALL: meaning condition must be true for all values in the set.

Example:

```

SELECT lname, fname
FROM employee
where salary > ALL (SELECT salary
                    FROM employee
                    where dno = 5);

```

EXISTS function

- Check whether the result of a nested query is empty or not.
- It is a Boolean function that returns a TRUE or FALSE.
- Could be used with NOT; NOT EXISTS

Example:

```

SELECT fname, lname
FROM employee
WHERE not EXISTS (SELECT *
                  FROM dependent
                  WHERE employee.ssn = dependent.ssn);

```

What does the query return?

How about this?

```

SELECT fname, lname
FROM employee
WHERE EXISTS (SELECT *
              FROM dependent
              WHERE employee.ssn = dependent.ssn)
AND
EXISTS (SELECT *
        FROM department
        WHERE employee.ssn = department.mgr_ssn);

```

A challenging example

Query: List first and last name of employees who work on ALL projects controlled by Dno=5.

```

SELECT      Fname, Lname
FROM        EMPLOYEE
WHERE       NOT EXISTS ( ( SELECT      Pnumber
                           FROM        PROJECT
                           WHERE       Dnum = 5)
                       EXCEPT      ( SELECT      Pno
                                       FROM        WORKS_ON
                                       WHERE       Ssn = Essn) );

```

EXCEPT operator is not implemented in mysql, there is a lab question asking you to rewrite this query such that you could run it in mysql. Hint: instead of EXCEPT use NOT IN, but you need to change the query a bit to make it work.

Based on the current database state in your computer (Workbench), the above query should return 0 rows. Try to write the query with NOT IN and for dnum=1. You should get the following records:

'Franklin','Wong'

'James','Borg'

'Jennifer','Wallace'

Explicit Sets

- You can use explicit set of values in WHERE clause.

```

SELECT DISTINCT essn
FROM works_on
where pno IN (1, 2, 3);

```

Renaming of Attributes in a query

- Use qualifier AS followed by a new name, to rename an attribute that appears in the result of a query.

```

SELECT E.Iname AS Employee_name, S.Iname as Supervisor_name
FROM employee as E, employee as S
WHERE E.super_ssn = S.ssn;

```

Joining Tables

- It allows you to join tables in the FROM clause.
- It is easier to comprehend, than specifying the table connections within WHERE clause.
- What this query retrieve?

```
SELECT fname, lname, address
FROM employee, department
where employee.dno = department.dnumber and department.dname = 'Research';
```

- Using join operator:

```
SELECT fname, lname, address
FROM (employee JOIN department ON employee.dno = department.dnumber)
where department.dname = 'Research';
```

Different types of join

- **INNER JOIN**
 - Default type of join in a joined table.
 - Tuple is included in the result only if a matching tuple exists in the other relation.
 - The join condition is specified explicitly. (As example Dno = Dnumber).
- **NATURAL JOIN on two relations R and S**
 - No join condition is specified.
 - R and S are joined on attribute(s) with the same name.
 - You may need to rename attributes of one relation so it can be joined with another using NATURAL JOIN.
 - Example:

```
SELECT fname, lname, address
FROM (employee NATURAL JOIN
      (department AS DEPT (dname, dno, mssn, msdate)))
WHERE dname='Research';
```

The implicit join condition is EMPLOYEE.Dno = DEPT.Dno.

We cannot run the above query, because renaming entire table like 'department AS DEPT (Dname, Dno, Mssn, Msdate)' is not supported in mysql.

- OUTER Joins

- LEFT OUTER JOIN

Every tuple in left table must appear in result.

If there is no matching tuple on right table, pad it with NULL values for attributes of right table.

Example:

Q8B:

```
SELECT E.lname AS Employee_name, S.lname as Supervisor_name  
FROM (employee as E LEFT OUTER JOIN employee as S ON E.super_ssn = S.ssn);
```

- RIGHT OUTER JOIN

Every tuple in right table must appear in result.

If no matching tuple on left table, pad it with NULL values for attributes of left table.